

Disclaimer: This document is meant for you to train for the final exam and check your knowledge about the topics in the MDK course. It is NOT meant to be (and will NOT be) a "fac simile" of the exam, since you will have to prove that you mastered the topics at a level in which you can apply the knowledge you gained to new problems and new questions. I suggest to try to solve these trainings after self study and confrontation with colleagues, to assess your level of preparation.

1 Multiple Choice Questions

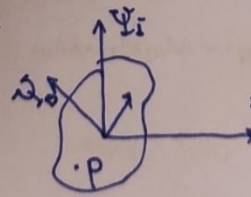
Read carefully. There is no limit in the number of right or wrong answers.

1. Since $so(3)$ is a Lie algebra, it possesses an antisymmetric operator called the commutator given by $[\tilde{\omega}_1, \tilde{\omega}_2] := \tilde{\omega}_1 \tilde{\omega}_2 - \tilde{\omega}_2 \tilde{\omega}_1$; $\tilde{\omega}_1, \tilde{\omega}_2 \in so(3)$. What is true about the commutator?

- ☒ $[\tilde{\omega}_1, \tilde{\omega}_2] \in so(3)$
☐ $[\tilde{\omega}_1, \tilde{\omega}_2] \in so^*(3)$
☐ $[\tilde{\omega}_1, \tilde{\omega}_2] = 0$ implies that either $\omega_1 = 0$ or $\omega_2 = 0$
☐ $[\tilde{\omega}, \tilde{\omega}] = 0$ if and only if $\tilde{\omega} = 0$
☐ The commutator of its arguments in tilde form represents the scalar product of its arguments in vector form.
☒ $[\tilde{\omega}_1, \tilde{\omega}_2] = 0$ implies that $\tilde{\omega}_1$ and $\tilde{\omega}_2$ commute.

2. Consider a rotating rigid body that is pinned from its center. Attached to the rotating body is the body-fixed frame Ψ_B . Now consider another fixed-intertial frame Ψ_I with the same origin as Ψ_B . The configuration of the body is described by R_B^I . Let p be a point attached to the rotating body. What is the velocity of p expressed in Ψ_I ?

- ☐ $\dot{p}^i = \omega_I^{B,B} \wedge p^i$
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$$\begin{aligned}
 p^i &= R_B^i p^B \Rightarrow \dot{p}^i = \dot{R}_B^i p^B + R_B^i \dot{p}^B \\
 &\Rightarrow \dot{p}^i = \tilde{\omega}_B^{ii} R_B^i p^B = \omega_B^{ii} \wedge p^i
 \end{aligned}$$

$\dot{R}_B^i = \tilde{\omega}_B^{ii} R_B^i$

3. Consider the Lie group $SE(3)$. What is true about the space $T_H^* SE(3)$, that is the cotangent space on $SE(3)$ at the configuration $H \in SE(3)$?

- ☐ There is a duality pairing between a twist $T \in se(3)$ and an element of $T_H^*SE(3)$, and the resulting scalar represents mechanical power along a rigid body motion.
- ☒ It is a 6-dimensional vector space.
- ☐ It has a natural inner product.
- ☒ An element of $T_H^*SE(3)$ can be naturally paired with the vector $\dot{H}$ and the resulting scalar represents mechanical power along a rigid body motion.
- ☐ It can be made into a Lie Group with a proper change of coordinates.
- ☒ It is the dual space of the space $T_HSE(3)$.

1. The matrix exponential is a map from $so(3)$ to $SO(3)$. If we want to obtain $R_A^B(t) = e^{Xt} R_A^B(0)$ what should be in the place of X ?

- ☒ $\tilde{\omega}_A^{B,B}$
- ☐ $T_A^{A,A}$
- ☐ $\tilde{T}_A^{B,A}$
- ☐ $\omega_B^{A,B}$
- ☐ W^b
- ☐ $\dot{H}_A^B$

By definition $\tilde{\omega}_A^{BB} = \dot{R}_A^B R_B^A$

$$\Rightarrow \dot{R}_A^B = \tilde{\omega}_A^{BB} R_A^B$$

$$\Rightarrow R_A^B(t) = e^{\tilde{\omega}_A^{BB} \cdot t} R_A^B(0)$$

2. The Euler-Lagrange equations in Robotics:

- ☐ are an alternative way to derive the equations with respect to the least action principle.
- ☒ are equations on the space of generalised forces on the configuration manifold T^*Q .
- ☐ are equations of the space of generalised velocities on the configuration manifold TQ
- ☒ Are n , with n the dimension of the configuration manifold Q , i.e. the DoFs of the Robot.

3. The Geometric Jacobian:

- ☐ Does not depend on the configuration of the robot.

- ☒ Contains the expression of the unit twists which characterise the kinematic pairs of the robot, expressed in the base frame Ψ_0 .
- ☐ has always a non null determinant if the robot is redundant.
- ☒ May loose rank in some configurations.
- ☐ Is a $(0 - 2)$ tensor.
- ☒ Represents a linear transformation from TQ to $se(3)$

4. The inertia tensor $\mathcal{I}$ of a rigid body:

- ☐ Is a linear transformation from $se(3)$ to $se(3)$
- ☒ Is a $(0 - 2)$ tensor.
- ☒ Is constant only in frames which are fixed with the rigid body.
- ☐ Is diagonal in any body fixed frame.
- ☐ Is constant only in the Principal Inertia Frame.
- ☒ In case the rigid body reduces to a point it just represents the mass of the resulting point mass.

2 Part 2: Open Question (weight=3)

1) Consider the dynamic equation (11) of a manipulator not touching the environment and ideally without friction, with mass matrix $M(q)$, $C(q, \dot{q})$ modelling the Coriolis and centrifugal forces, and $G(q)$ the gravitational forces due to the potential energy $V(q)$ and τ are the torques applied on the joints:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau \quad , \quad G(q) := \frac{\partial V}{\partial q}(q) \quad (11)$$

How would the equation (11) change if a linear viscous friction with constant damping coefficient $b_i > 0$ is present at joint i , with $1 \leq i \leq n$? And how would it change in case an external wrench W is applied to the end effector, and expressed numerically in the end-effector frame as W^n ? Motivate your answer.

2) Describe the relation between a change of coordinates between two frames (which we normally represent through an homogeneous matrix) and an isometric motion on the 3-D Euclidean space.

OPEN QUESTIONS

1) a) On each joint the generalised force τ_i is
 $\tau_i = -b_i \dot{q}_i$ (viscous friction).

So as a vector $\tau = -B \dot{q}$ where $B = \text{diag}\{b_i\}$

$$\Rightarrow \boxed{M(q)\ddot{q} + C(q, \dot{q})\dot{q} + B\dot{q} + G(q) = 0}$$

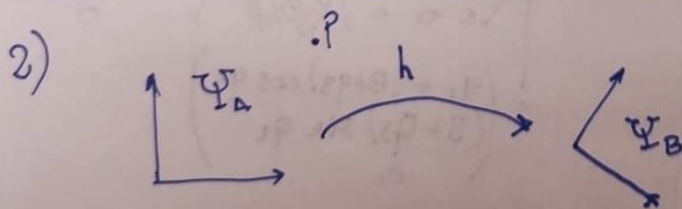
b) Since $T_n^{0,0} = J(q)\dot{q}$, by duality

$\tau = J^T(q)W^0$ where τ are the generalised forces at the joints, $J(q)$ is the geometric jacobian and W^0 is the wrench acting on the end-effector expressed in Ψ_0 (inertial frame).

If W^n is the wrench in the end-effector frame,

$W^0 = \text{Ad}_{H_0^n}^T W^n$. So the model becomes

$$\boxed{M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = J^T(q) \text{Ad}_{H_0^n}^T W^n}$$



For a fixed point P , $P^A = H_B^A P^B$ (change of coordinates from Ψ_B to Ψ_A)

A motion h is such that

$\varphi_B \circ h \circ \varphi_A^{-1} = I$, where φ_B maps P to P^B and φ_A maps P to P^A .
 (i.e., $H_B^A = \varphi_A \circ \varphi_B^{-1}$). But $\varphi_B \circ \varphi_A^{-1} \circ \varphi_A \circ h \circ \varphi_A^{-1} = I$ as well,
 and $H_A^B = \varphi_B \circ \varphi_A^{-1}$, so $\boxed{\varphi_A \circ h \circ \varphi_A^{-1} = H_A^B} \Rightarrow$ The motion h expressed in Ψ_A is the inverse of the change of coord. from Ψ_A to Ψ_B

3) No square matrix $\Rightarrow$ no determinant!

[P.S. I become violent if you talk about determinant of non square matrices :)]

~~Exercise~~

4) I start by solving 2), i.e., calculating the geometric jacobian $J(q)$ such that $T_4^{00} = J(q) \dot{q}$.

$J(q) = \begin{pmatrix} \hat{T}_1 & \hat{T}_2 & \hat{T}_3 & \hat{T}_4 \end{pmatrix}$ where $\hat{T}_i$ are the unit twists in $\mathcal{V}_0$ corresponding to i -th joint.

$$\hat{T}_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\hat{T}_2 = \begin{pmatrix} \omega_2 \\ \pi_2 \wedge \omega_2 \end{pmatrix}, \quad \omega_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\pi_2 \wedge \omega_2 = -\tilde{\omega}_2 \pi_2 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ q_1 \\ 0 \end{pmatrix} = \begin{pmatrix} q_1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \hat{T}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ q_1 \\ 0 \end{pmatrix}$$

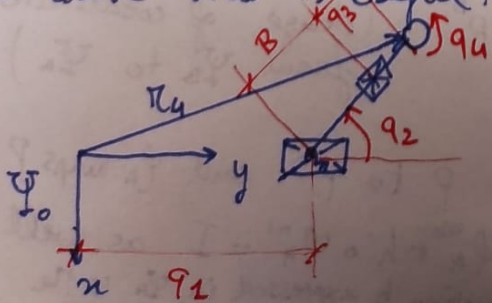
$$\hat{T}_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -\sin q_2 \\ \cos q_2 \\ 0 \end{pmatrix}$$

$$\hat{T}_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ q_2 + (B+q_3) \cos q_2 \\ (B+q_3) \sin q_2 \\ 0 \end{pmatrix}$$

$$\hat{T}_4 = \begin{pmatrix} \omega_4 \\ \pi_4 \wedge \omega_4 \end{pmatrix}, \quad \omega_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\pi_4 \wedge \omega_4 = -\tilde{\omega}_4 \pi_4 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ q_1 + (B+q_3) \cos q_2 \\ 0 \end{pmatrix} = \begin{pmatrix} q_1 + (B+q_3) \cos q_2 \\ (B+q_3) \sin q_2 \\ 0 \end{pmatrix}$$

From above view is useful:



$$\pi_4 = \begin{pmatrix} 0 \\ q_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -(B+q_3) \sin q_2 \\ (B+q_3) \cos q_2 \\ 0 \end{pmatrix}$$

In order to solve 1) we need the twists in reference config. $\Rightarrow$ It suffices to evaluate $\hat{T}_i$ at $q=0$
 i.e. $\hat{T}_i^{\text{ref}} = \hat{T}_i|_{q=0}$ (It also provides a good check!)

and then

$$H_u^0(q) = \sum_{i=1}^4 e^{\hat{T}_i^{\text{ref}} q_i} \cdot H_u^0(0)$$

where $H_u^0(0) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & B \\ 0 & 0 & 1 & (A-C) \\ 0 & 0 & 0 & 1 \end{pmatrix}$

3) $\dot{E}^0 = H_u^0 \dot{E}^4$

$\dot{E}^0 = H_u^0 \dot{E}^4$ since $\dot{E}^4 = 0$

$\dot{H}_u^0 = \tilde{T}_u^{00} H_u^0 \Rightarrow \dot{E}^0 = \tilde{T}_u^{00} H_u^0 \dot{E}^4 \Rightarrow \dot{E}^0 = \underbrace{J(q)}_{\text{from 1)} } \dot{q} \underbrace{H_u^0(q)}_{\text{from 2)} } \dot{E}^4$

4) $\mathcal{L}^T = J^T(q) (W^0)^T$

$(W^0)^T = \text{Ad}_{H_u^0}^T (W^4)^T$

$\hookrightarrow$ inverse matrix of $H_u^0(q)$, from 2)

$$\Rightarrow \boxed{\mathcal{L}^T = J^T(q) \text{Ad}_{H_u^0}^T (W^4)^T}$$